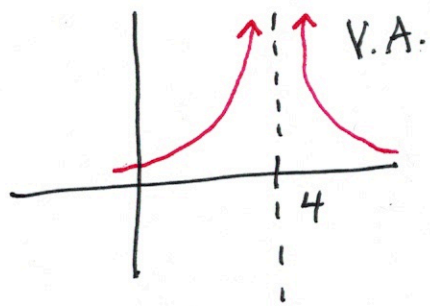
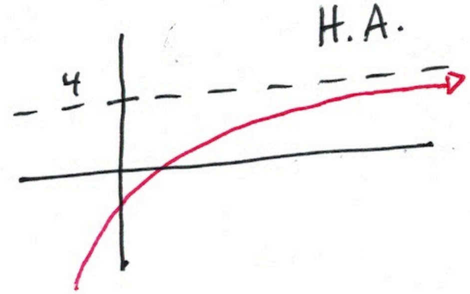
Section 2.6 Limits Involving Infinity





$$\lim_{x \to 4} f(x) = \infty$$

$$\lim_{x \to \infty} f(x) = 4$$

Ex ① $\displaystyle \lim_{x \to 4} \frac{x^2-16}{x-4} = \lim_{x \to 4} \frac{\cancel{(x-4)}(x+4)}{\cancel{x-4}} = 4+4 = \boxed{8}$

② $\displaystyle \lim_{x \to 4^+} \frac{x^2-16}{(x-4)^2} = \lim_{x \to 4^+} \frac{\cancel{(x-4)}(x+4)}{\cancel{(x-4)}(x-4)} = \lim_{x \to 4^+} \frac{x+4}{x-4} = \frac{8}{0} \to \infty$

But on the left, the limit is $-\infty$.

$\displaystyle \lim_{x \to 4} \frac{x^2-16}{(x-4)^2}$ DNE.

Ex 3 $$\lim_{x\to\infty} \frac{x^2-16}{(x-4)^2} = \lim_{x\to\infty} \frac{(x^2-16)\frac{1}{x^2}}{(x^2-8x+16)\frac{1}{x^2}}$$

Find the most powerful term, x^2.

Divide every term by x^2.

$$\lim_{x\to\infty} \frac{\frac{x^2}{x^2} - \frac{16}{x^2}}{\frac{x^2}{x^2} - \frac{8x}{x^2} + \frac{16}{x^2}}$$

$$\lim_{x\to\infty} \frac{1 - \cancel{\frac{16}{x^2}} \to 0}{1 - \cancel{\frac{8}{x}} \to 0 + \cancel{\frac{16}{x^2}} \to 0}$$

$$= \frac{1}{1} = \boxed{1}$$

Ex ④ $\lim_{x \to \infty} \dfrac{\sqrt{x^2+9}}{2x+8} \approx \dfrac{\sqrt{x^2}}{2x} = \dfrac{x}{2x} = \dfrac{1}{2}$ ✓ ☺

$\sqrt{9x} = 3\sqrt{x}$

$$\lim_{x \to \infty} \frac{\frac{1}{x}\sqrt{x^2+9}}{\frac{1}{x}(2x+8)}$$

$$\lim_{x \to \infty} \frac{\sqrt{\frac{x^2}{x^2}+\frac{9}{x^2}}}{\frac{2x}{x}+\frac{8}{x}}$$

$$\lim_{x \to \infty} \frac{\sqrt{1+\frac{9}{x^2}\to 0}}{2+\frac{8}{x}\to 0}$$

$$= \frac{\sqrt{1}}{2} = \boxed{\frac{1}{2}}$$

Ex 5 $\lim_{x \to \infty} \dfrac{42}{1 + 5e^{-x}}$

$$\lim_{x \to \infty} \frac{42}{1 + \frac{5}{e^x} \to 0} = \frac{42}{1} = \boxed{42}$$